



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 03.08.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No : PHA G 617
Course Title : Advanced Drug Delivery Systems
Instructor-in-Charge : Prof. Murali Monohar Pandey
Instructor(s) : NA
Tutorial/Practical Instructors:

1. Course Description:

A study of physicochemical and biopharmaceutical factors involved in the design of novel drug delivery systems like mucosal, particulate systems for systemic delivery of bioactive molecules. Special considerations for delivery of protein, peptide and other biological products. In vitro and in vivo evaluation of novel drug delivery systems

2. Scope and Objective of the Course:

The prime objective of this course is to impart knowledge of design, development and evaluation of novel drug delivery systems (NDDS). The primary focus would be on integrating the biopharmaceutical, physiochemical and physiological properties for design and development of (NDDS) with due consideration of pharmacological action. Excipients used in design and development of NDDS would be covered exhaustively.

3. Text Books:

- i. Tyle, P. Specialized Drug Delivery Systems- Manufacturing and Production Technology, Marcel Dekker, New York, 1990

4. Reference Books:

- i. Prescott, L.F., and Nimmo, W.S. Novel Drug Delivery, John Wiley & Sons, Chichester, 1989.
- ii. McNally, E. J. Protein Formulation and Delivery, Marcel Dekker, New York, 2000.
- iii. Frokjaer, S., and Hovgaard, L. Pharmaceutical Formulation Development of Peptides and Proteins, Taylor and Francis, London, 2000.
- iv. Nail and Akers, Development and manufacture of protein pharmaceuticals, Springer, 1st Edition, 2009

5. Course Plan:

Module Number	Lecture session/Tutorial Session.	Reference	Learning Outcome
1. Overview of NDDS, Opportunities and challenges	L1.1. Drug discovery Vs. product development, Conventional formulations, Why Novel drug delivery systems?,	T.1 CH.1 Class notes	Understanding of NDDS and alternate routes of administration
	L 1.2. Overview on NDDS and alternate routes of administration	T.1 CH.1 Class notes	
2. Various aspects affecting design, development and selection of NDDS	L2.1. Physicochemical, Biopharmaceutical and Physiological factors important for design of NDDS	R.1 CH. 2 & 7, class notes	Understanding the relationship between Physicochemical,



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	L2.2. 4D approach, Case study	R.1 CH. 2 &7, class notes	Biopharmaceutical and Physiological factors for design of suitable delivery system
3. Design, characterization and evaluation of advanced drug delivery systems	L3.1-L3.2. Techniques used for development of NDDS	R.1 CH. 2 &7, class notes	Understanding of materials, method of preparations, advantages and disadvantages and techniques for development of NDDS
	L3.3.-L3.10. Nanosuspensions, Microparticles, Nanoparticles, Liposomes, Polymeric micelles, Polymer drug conjugates, Dendrimers	T.1 CH.5, Journal articles, class notes	
4. Targeted Drug delivery	L4.1 Various drug delivery systems for site specific targeting,	R.1 CH.9,25,&32	Understanding the passive and active targeting strategies. EPR effect. First, second and third order targeting.
	L4.2.-L4.3. Design parameters for tumor targeted delivery of nanotherapeutics.	R.1 CH.9,25,&32	
5. Oral delivery of nano-therapeutics	L5.1. Basic design considerations,	Class notes, Journal articles	Understanding on oral delivery of NDDS.
	L5.2.-L5.4. Mechanism of uptake via oral route, different nano-carrier systems for oral delivery	Class notes, Journal articles	
6. Pulmonary delivery of nano-therapeutics	L6.1. Basic design considerations,	Class notes, Journal articles	Understanding on pulmonary delivery of NDDS.
	L6.2.-L6.4. Mechanism of uptake via pulmonary route	Class notes, Journal articles	
7. Drug delivery of proteins and peptides	L7.1.-L7.6 Basic considerations in the design of Protein/Peptide based delivery systems	R4. Ch-1 and Ch 2 R3 ch 4	Basics of protein and peptides delivery, How they are different from small molecules, Design considerations.



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8. NanoRNAi	L 8.1 Introduction to siRNA, miRNA. Gene therapy	Class notes, Journal articles	Understanding on siRNA, miRNA. Polymeric systems used for their Delivery
	L 8.2.-L8.3 Nano-carriers for RNAi		
9. In vitro Characterization	L 9.1-L 9.2. Various methods and techniques for in-vitro drug release and other characterization of NDDS	Class notes, Journal articles	Methods for in vitro assessment of NDDS
10. In vivo Characterization and Evaluation	L10.1-10.4 Selection of appropriate animal models for evaluation of various NDDS	Class notes, Journal articles	Methods for in vivo assessment of NDDS

6. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Nature of component (Close Book/ Open Book)
Mid-Semester Test	90 Min.	30		CB/OB
Lab component/ Literature Survey, Seminars/ Research Summaries/Assignments/Quizzes		35		
Comprehensive Examination	3 h	35		CB/OB

*Slides used during class hours provide key information for which additional supportive information is expected to be collected from sources aforementioned. Recent developments in the area/topic will be discussed in class based on their significance to healthcare delivery and hence some information on therapeutic benefits and toxicity effects, besides others, may differ from the information in text, reference material and hence students are expected to take note of such key discussions during contact hours. Such discussions held in class will be considered as primary source of information in assessments.

7. Consultation Hour: To be announced in the class.

8. Notices: All the notices pertaining to this course will be displayed on Pharmacy noticeboard/Nalanda.

9. Make-up Policy: Make-ups are not given as a routine. It is solely dependent on the “genuineness” of the circumstances under which a student fails to appear in a scheduled evaluation component. Prior permission should be sought from the instructor-in-charge in advance.



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10. Note (if any): Grading Procedure: As specified in Handout – Part I, appended to the timetable, the instructor in-charge reserves the right to award a NC report in case the student does not make himself/ herself available for any of the evaluation component mentioned above. Also it is not imperative on part of the instructor in-charge to award all the grades. Borderline cases during grading will be judged on the basis of regularity to classes and consistency or progress in the performance in evaluation components. The maximum pull-up to be exercised by the instructor in-charge will be announced in the class and shall be based on the subjective judgment of the evaluator.

All evaluation components are equally important, irrespective of weightage. Hence, students failing to attend scheduled classes, or absenting themselves in one or many of the evaluation components, may become ineligible for obtaining a valid grade at the end of the semester. Attendance in lectures and practicals are all equally important as they are all integral components of learning, irrespective of weightage and may be taken into consideration, during grading.

Hence, students are strongly advised to keep away from absenting themselves from all aforementioned contact sessions. Clearing the course would require adequate performance in quizzes, tests**, examinations**, and in practical (laboratory) components, separately (i.e. procuring low marks in evaluation components, aforementioned**, other than laboratory practicals, would not suffice, to clear the course).**

Instructor-in-charge
Course No. PHA G617